

Verifier-Gated LLM NetOps: A Preregistered, Artifact-Backed Evaluation of Input-Sanitization and Deterministic Verification Against Adversarial Intent

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Abstract—We report a preregistered paired confirmatory experiment (study_id cand-2026-03) that measured whether template-based input sanitization combined with deterministic verifier-gating (and at-most-one automated repair) changes the rate at which a deterministic validator accepts LLM-generated CLI configurations under adversarially mutated intents. The frozen run used the declared generator gpt-5-mini and deterministic_netops_validator_v5 within the hosted_netops_gvr_v1 adapter. Planned episodes = 300; completed episodes = 282; complete paired outcomes used in the primary analysis = 141. Observed per-pair acceptance: Baseline 135/141 (95.7447%); Guarded 141/141 (100.0%); paired absolute difference = 0.04255319148936165; exact two-sided McNemar $p = 0.03125$; 95% bootstrap percentile CI (10,000 resamples, seed = 1729) = [0.014184397163120567, 0.07801418439716312]. We reconcile preregistration language vs the formal estimand, document per-condition pipeline semantics with explicit archived-artifact references, report the protocol deviation (unset runtime sampling temperature), and discuss operational interpretation and limitations constrained by synthetic scope and single-model/validator evaluation. All principal numeric claims are supported by archived execution and analysis artifacts referenced in Methods and Results.

I. INTRODUCTION

Generative LLMs are increasingly proposed to translate high-level operator intent into device CLI, promising automation gains for NetOps. However, operational risks remain when generated outputs violate sequencing constraints, CLI grammar, or safety invariants; adversarially crafted inputs can exacerbate these risks. Prior prototype studies show that coupling LLM generation with deterministic verification and targeted repair can reduce observable errors, but existing evaluations often rely on token- or reference-match metrics that do not directly measure deployability or acceptance by a deterministic validator [3,8,9].

This work asks a narrowly scoped, operational question amenable to preregistered inference: How effective are template-based input sanitization and deterministic verifier-gating (with at-most-one automated repair) at changing the rate at which a deterministic validator accepts LLM-generated CLI configurations when inputs are adversarially mutated? The study cand-2026-03 preregistered a single formal estimand (paired_success_rate_difference_guarded_minus_baseline) and a confirmatory paired test. Crucially, because the

registered generation semantics were shared_initial_candidate, the paired comparison isolates downstream guarded-pipeline effects (sanitization, gating, permitted repair, and final validation) applied to a single shared LLM draft per intent rather than differences in initial generation.

Contributions. (1) A preregistered, artifact-backed paired experiment executed end-to-end and reconciled against the archived preregistration; (2) a precise, artifact-grounded description of per-condition pipeline semantics that demonstrates shared_initial_candidate reuse; (3) quantitative confirmatory results showing a small but statistically detectable absolute change in deterministic-validator acceptance under the guarded pipeline on a synthetic adversarial-intent bank; and (4) an explicit reconciliation of preregistration natural-language hypothesis wording with the formal estimand and a disclosure of a protocol deviation (unset sampling temperature) together with reproducibility guidance.

II. RELATED WORK

Deterministic network-verification tools. Formal, deterministic validators such as Header Space Analysis and VeriFlow (and related NetPlumber-style approaches) are widely used to encode and rapidly check forwarding, access-control, and sequencing invariants before deployment [5–7]. These tools operate over explicit, analyzable models of packet headers or configuration effects, and therefore directly report property-level violations (reachability, isolation, rule precedence) rather than syntactic or token-level differences. That representational difference explains their suitability as deployment gates in LLM-assisted NetOps: validators can accept or reject a candidate configuration with a semantics-focused criterion, but they require explicit formalization of the intended invariants and do not automatically cover every dynamic or protocol-level safety property. Prior literature therefore treats deterministic validators as precise but partial oracles whose effectiveness depends on the fidelity of the encoded specifications and the scope of checked properties [5–7,14].

LLM-assisted configuration and evaluation gaps. Recent empirical work that studies LLMs for NetOps (e.g., NetConEval, NetLLMBench, and zero-touch proposals) consistently finds that model-only generation produces both syntactic errors

and higher-level semantic mistakes (ordering, precondition violations, protected-interface breaches) when translating intent to CLI [3,8,9]. Many published evaluations focus on string- or reference-match metrics that measure surface similarity to a gold command sequence; such metrics correlate poorly with deployability in practice because they neither verify execution-order constraints nor check reachability/forwarding invariants. Surveys of AIOps/telecommunications emphasize this fragmentation in evaluation practices and call for operationally grounded benchmarks that measure deployability, safety, and validator acceptance rather than token overlap alone [4,9].

Generate-validate-repair architectures and tool-assisted orchestration. A common mitigation pattern is the generate-validate-repair architecture: an LLM produces a candidate, an automated verifier checks it, and feedback is converted into constrained repair prompts or corrective actions that produce revised candidates. Prototype studies and orchestration patterns (including Toolformer-style approaches) demonstrate that integrating external checks into the generation loop improves concrete outcome metrics on narrow tasks, provided verifier feedback is sufficiently precise and repair scope is constrained [11,3]. These demonstrations show methodological promise but also reveal dependencies: (a) repair effectiveness depends on the granularity and clarity of verifier diagnostics (e.g., whether the checker returns a parsable, actionable violation code vs a high-level failure label), and (b) bounded repair policies (limited number of automatic repair attempts and explicit constraints on allowable edits) are often necessary to avoid uncontrolled changes that could mask adversarial manipulations.

Security and adversarial inputs in LLM-integrated NetOps. The LLM-systems security literature documents prompt-injection and indirect prompt-attacks that can subvert downstream behavior when external or user-supplied text is trusted by an orchestrator [10]. These attack modes motivate operational defenses used in practice, such as template-based input sanitization, slot validation, and gating designs that deterministically block candidates failing explicit safety checks. Prior defensive evaluations are often performed on synthetic or constrained datasets and may not exhaustively explore multi-source or retrieval-augmented pipelines, leaving an open need for NetOps-specific threat-modeling and rigorous mitigation studies.

Benchmarks, measurement, and operational desiderata. Several benchmark efforts and proposals (NetConfEval, NetLLM-Bench, and related task suites) attempt to exercise common NetOps intents but differ in goals: some emphasize generation diversity, others target protocol-specific behaviors. Across these efforts a recurring limitation is the lack of standardized, validator-centered deployability metrics. This paper situates itself against those gaps by preregistering an estimand that uses deterministic-validator acceptance as the primary quantity of interest and by explicitly using `shared_initial_candidate` semantics so that post-generation pipeline elements (sanitization, gating, bounded repair) are the causal factors under test. Unlike many prior demonstrations, the present study em-

phasizes artifact-backed traceability (shared-generation files, per-episode validator traces, and provider-call accounting) to connect observed numeric differences to concrete pipeline actions and to diagnose discordant cases traceable to permitted repairs.

Synthesis and remaining limitations in prior work. In sum, prior work establishes three relevant points: (1) deterministic validators provide semantically meaningful acceptance criteria but require explicit property specification; (2) generate-validate-repair architectures can reduce observable errors but their safety depends on verifier diagnostic quality, repair scope, and gating policies; and (3) existing evaluations often use token-level metrics or nonstandardized benchmarks that do not directly measure deployability under an explicit danger model. These findings motivate the present preregistered, artifact-backed paired experiment that isolates post-generation safeguards under `shared_initial_candidate` semantics and records per-episode validator traces to support interpretive diagnostics.

III. METHODOLOGY

Preregistration and estimand (artifact grounding). The study `cand-2026-03` was preregistered and archived prior to observing outcomes (preregistration_sha256: `ba6ed25e84c7240f94f2d50b3e778fbfffd810ff2db632df8946e85f0f1373`). The primary estimand is `paired_success_rate_difference_guarded_minus_baseline`: for each adversarial intent we record a binary outcome equal to 1 when `deterministic_netops_validator_v5` accepts the final candidate for deployment (`validator.validation_after.valid == true` and `violation_count == 0` in scoring artifacts) and 0 otherwise. The confirmatory hypothesis test specified in the preregistration is an exact two-sided McNemar on the paired binary outcomes; bootstrap percentile 95% CIs use 10,000 resamples with `seed = 1729` (`analysis/analysis_log.jsonl`).

Design, task bank, and adversary model. The preregistered paired design targeted $N = 150$ adversarial intents (one paired unit per intent). Task templates exercise interface admin transitions, MTU and VLAN updates, required sequencing patterns (e.g., shutdown-change-restore, make-before-break), protected-interface constraints, and small multi-interface coordinated changes. Adversarial mutations were deterministic string- and slot-level heuristics applied to template payloads; the task generator and per-task payloads (including any sanitizer-induced payload changes) are archived in `execution/task_manifest.jsonl` (task generator id `deterministic_netops_task_generator_v5`). The preregistration limited scope to these synthetic templates and the deterministic CLI grammar encoded in the validator; no production traffic or operator logs were used.

Model, validator, and orchestration (explicit recorded configuration). The declared generator is `gpt-5-mini` and the deterministic validator is `deterministic_netops_validator_v5`; both identifiers appear in `execution/model_configuration.json` and in per-episode scoring artifacts (e.g., `execution/scoring/task-000001-baseline.json`). The orchestration adapter family

hosted_netops_gvr_v1 implements the generate-validate-repair pattern used in this run. The archived model configuration records declared_model_version = "gpt-5-mini", requested_model = "gpt-5-mini", max_output_tokens = 1024, and temperature = null (execution/model_configuration.json). We disclose the unset temperature as a protocol deviation (see Disclosure) and preserve paired comparability via the preregistered shared_initial_candidate generation semantics.

Per-pair pipeline semantics and explicit artifact-backed timeline. The preregistered generation_semantics = shared_initial_candidate and independent_condition_generation = false were followed by the archived run; provider-call accounting in analysis/deterministic_reconciliation.json shows stage_counts.shared_initial_generation = 150 and stage_counts.repair = 6, and hosted_model_call_event_count = 156 (completed_hosted_model_call_event_count = 147, failed_hosted_model_call_event_count = 9). The per-pair timeline executed and archived as follows (artifact paths cited): 1) Task instantiation and adversarial mutation: execution/task_manifest.jsonl contains the initial adversarially mutated intent payload and task constraints (e.g., allowed_touched_fields, workflow_policies). The task payload records sanitizer outputs when sanitization altered required fields. 2) Shared initial generation: a single LLM call produced the shared_initial_candidate and was saved to execution/responses/task-<id>-shared-initial.txt. Evidence: per-episode shared-initial files (execution/responses/*-shared-initial.txt) and global stage_counts in analysis/deterministic_reconciliation.json. 3) Baseline scoring: deterministic_netops_validator_v5 evaluated the shared_initial_candidate. Baseline scoring artifacts (execution/scoring/task-<id>-baseline.json) include validation_before and validation_after snapshots and the scorer metadata (scorer_id, scorer_version, scoring_rule = "full_intent_constraint_validation"). Representative examples: execution/scoring/task-000001-baseline.json, execution/scoring/task-000002-baseline.json. 4) Guarded branch processing: the guarded branch applies a template-based input-sanitizer prior to gating. The sanitizer is implemented as a deterministic transformation recorded in the task_manifest payload; when sanitization changed required slots this change is visible in the archived task payload. The verifier-gate evaluates the sanitized candidate. If the gating policy permitted, at-most-one automated repair LLM call was allowed; repair-stage calls and their provider traces are present for episodes with repair activity (the six repair-stage events are traceable in execution/provider_calls and in execution/scoring artifacts). Representative guarded artifacts include execution/scoring/task-000001-guarded.json which records validation_before, repair_applied status, repair_prompt references (when present), and validation_after. 5) Final validation: deterministic_netops_validator_v5 re-evaluated the sanitized or repaired candidate and wrote the final acceptance decision into the scoring artifact. The binary outcome for the estimand is derived directly from the

validator's validation_after.valid field and violation_count == 0.

Operationalization of success/failure, scoring rules, and deterministic checks. The scoring rule used for the primary outcome is full_intent_constraint_validation implemented by deterministic_netops_validator_v5; this validator enforces required-sequence constraints, allowed/touched field restrictions, protected-interface invariants, and CLI grammar. A successful outcome for a condition is defined as the validator marking the final candidate valid with zero violations (as visible in the validation_after object in execution/scoring/*.json). The paired-contingency counts are taken from analysis/paired_contingency_table.csv where rows and columns correspond to guarded and baseline binary outcomes respectively.

Missingness, technical failures, exclusions, and preregistered handling. The preregistered missingness_plan excluded a paired unit only when both paired runs were technical failures (model API failure, verifier timeout, or parser error). The execution manifest reports planned_episode_count = 300, completed_episode_count = 282, and failed_episode_count = 18 (execution/execution_manifest.json). The analysis missingness summary (analysis/missingness_summary.csv) records complete_pairs = 141 and both_missing_pairs = 9, consistent with the preregistered exclusion rule. Provider-call accounting in analysis/deterministic_reconciliation.json documents hosted_model_call_event_count = 156, completed = 147, failed = 9, and stage_counts (shared_initial_generation = 150, repair = 6), which together explain the observed missingness and repair incidence.

Confirmatory and supplementary analysis procedures (archive-referenced). The primary confirmatory test is an exact two-sided McNemar on the paired binary outcomes; this test and the paired point estimate (guarded - baseline) are computed by the archived paired_binary_analysis_v1 executor (analysis/results.json). Confidence intervals are bootstrap-percentile with 10,000 resamples and seed = 1729; bootstrap parameters and the completed-analysis log are archived in analysis/analysis_log.jsonl. Discordant-case diagnostics (n10 and n01 pairs) were inspected by tracing per-episode scoring artifacts referenced in analysis/paired_contingency_table.csv and execution/scoring/*.json to determine whether repairs or sanitization accounted for acceptance changes; the six discordant n10 pairs are associated with repair-stage activity recorded in execution/provider_calls and scoring artifacts.

Preplanned secondary/sensitivity checks and unexecuted items. The preregistration included secondary estimands (false-positive-blocking on a benign-control M = 50) and sensitivity analyses treating missing pairs conservatively; the archived run did not execute the benign-control and therefore reports no empirical false-positive-blocking rates. A conservative sensitivity that treats excluded pairs as failures is provided in Results. The preregistration also specified multiplicity handling (single primary confirmatory test; exploratory reporting with BH-FDR for families of secondary tests) and a contamination/embargo plan for hazardous arti-

facts; artifact-level contamination summaries are recorded in `analysis/contamination_summary.csv`.

Reproducibility notes (artifact pointers and protocol deviation). Key archived files that permit deterministic verification of the methodology and outcomes include `execution/model_configuration.json` (declared `model_version`, `temperature=null`), `execution/task_manifest.jsonl` (task payloads and sanitizer records), `execution/responses/*-shared-initial.txt` (shared initial candidates), `execution/scoring/*.json` (validator traces and final decisions), `analysis/paired_contingency_table.csv`, `analysis/condition_summary.csv`, `analysis/analysis_log.jsonl`, `analysis/deterministic_reconciliation.json`, and `execution/execution_manifest.json`. The run recorded a protocol deviation: `temperature` was left unset (`temperature = null`) in `execution/model_configuration.json`. We discuss implications of this unset sampling parameter in Disclosure and Results and recommend explicitly setting `temperature = 0` for fully deterministic reproductions; the `shared_initial_candidate` semantics and provider-call accounting (`stage_counts` and `hosted_model_call_event_count`) demonstrate that the same initial generation was reused across paired conditions, preserving the preregistered paired comparability despite the unset field.

IV. RESULTS

Execution and counts. The frozen run produced planned episodes = 300, completed episodes = 282, failed episodes = 18, and model provider calls used = 156 (`execution/execution_manifest.json`). The primary confirmatory analysis used 141 complete pairs (`analysis/missingness_summary.csv`). Condition summary (`analysis/condition_summary.csv`) shows `baseline_successes` = 135/141 (0.9574468085) and `guarded_successes` = 141/141 (1.0).

Paired contingency and inference. The archived paired contingency (`analysis/paired_contingency_table.csv`) is `n11` = 135, `n10` = 6 (`guarded`=1, `baseline`=0), `n01` = 0, `n00` = 0. An exact two-sided McNemar test on the observed paired outcomes yields $p = 0.03125$. The point estimate (`guarded` - `baseline`) = 0.04255319148936165. The 95% bootstrap percentile CI (10,000 resamples, seed = 1729) is [0.014184397163120567, 0.07801418439716312]; bootstrap parameters and log entries are archived in `analysis/analysis_log.jsonl`.

Discordant-case diagnostics. The six discordant `n10` pairs are fully traceable in the archive and are associated with repair-stage activity: `analysis/deterministic_reconciliation.json` records repair = 6 and the corresponding per-episode scoring artifacts contain validator before/after snapshots showing sanitizer and/or permitted repair changes that converted a rejected initial draft into an accepted final candidate. Representative artifacts include `execution/responses/task-000001-shared-initial.txt`, `execution/scoring/task-000001-baseline.json` and `execution/scoring/task-000001-guarded.json`; these artifacts explicitly show the identical `shared_initial_candidate`

referenced by both scoring artifacts and the validator’s `validation_before` and `validation_after` objects.

Missingness sensitivity and unexecuted items. Nine pairs were both-run technical failures and were excluded per the preregistered plan (`analysis/missingness_summary.csv`: `both_missing_pairs` = 9). Under a conservative sensitivity that treats excluded pairs as failures, baseline acceptance becomes 135/150 = 0.9000 and guarded acceptance 141/150 = 0.9400. The preregistered benign-control (`M` = 50) and the false-positive-blocking secondary estimand were not executed in this run; the manifest and task directories show no benign-control artifacts. Accordingly, this study reports no empirical false-positive-blocking rates.

V. DISCUSSION

Hypothesis reconciliation and interpretive boundary. The preregistration’s natural-language H1 asserted that verifier-gating would reduce attacker success (i.e., fewer unsafe configurations accepted). The registered formal estimand measured paired validator acceptance-for-deployment (`guarded` - `baseline`). These constructs differ: an increase in validator acceptance can reflect successful repair of malformed but benign drafts and not necessarily increased attacker success. Our observed `guarded`→`higher` acceptance (≈ 4.255 percentage points) occurred because sanitizer and permitted repair recovered repairable drafts that the baseline validator rejected. Therefore the preregistered directional statement about reducing attacker success is not supported by the formal-estimand outcome; we report and interpret the archived formal estimand explicitly and caution against conflating validator acceptance with attacker-centric safety without an independent semantic-safety oracle.

Operational implications and repair-gate co-design. Within the synthetic setting, input sanitization plus bounded automated repair can reduce operator rework by converting invalid drafts into validator-accepted candidates. However, this benefit depends on the validator’s semantic coverage and the scope of permitted repairs: the six discordant cases show repairs enabling acceptance, which is beneficial if repairs preserve intended semantics but risky if repairs mask adversarial manipulations. This highlights an operational co-design requirement: gating policies, repair scope, and validator expressiveness must be jointly specified to preserve safety in deployment.

Threats to validity. Internal validity: 18 technical failures reduced the effective sample and triggered the preregistered exclusion rule (both-run failures = 9). Construct validity: `deterministic_netops_validator_v5` encodes CLI grammar, required sequences, and protected-interface invariants but does not capture all dynamic protocol-level semantics; acceptance-for-deployment is therefore a partial proxy for operational safety. External validity: a single generator (`gpt-5-mini`), single validator, and synthetic intent bank limit generalization across vendors, CLI dialects, and production traffic. Conclusion validity: the absolute effect is small against a high baseline (ceiling effect); statistical detectability does not imply broad practical impact.

Reproducibility guidance and protocol deviation. The archived execution/model_configuration.json recorded temperature = null (unset). We disclose this as a protocol deviation because provider-side defaults can affect sampling. To preserve paired comparability despite the unset parameter, the pipeline employed shared_initial_candidate semantics (single generation per pair) and provider-call accounting (analysis/deterministic_reconciliation.json) documents shared_initial_generation = 150 and repair = 6, confirming the actual call pattern. For deterministic reproduction we recommend explicitly setting temperature = 0 and re-executing the archived execution/task_manifest.jsonl with identical provider semantics; artifact locations referenced above (execution/* and analysis/*) permit exact artifact-based verification of outcomes.

Recommendations. Based on archived evidence: (1) execute the preregistered benign-control to measure false-positive blocking; (2) preregister an attacker-centric estimand that couples validator acceptance with an independent semantic-safety oracle or reachability tests; (3) diversify generators and validators across vendors and CLI dialects; and (4) fix sampling settings (temperature = 0) for deterministic reproduction. Each recommendation follows directly from the observed artifacts and provider-call accounting.

VI. LIMITATIONS

- Synthetic task bank and intent templates — not real production traffic or operator logs; results may not generalize to field datasets.
- Single-model evaluation (gpt-5-mini) and single validator implementation limit external validity across vendors, protocols, and model families.
- Validator coverage constrained to encoded safety properties (CLI grammar, required sequences, protected-interface invariants); some protocol-level or dynamic runtime semantics are not represented.
- Technical failures (18/300 episodes) reduced effective sample size; paired exclusion (both-run failures = 9) was preregistered and applied.
- Adversary model limited to mutations of synthetic intents; prompt-injection in retrieval-augmented or multi-source pipelines was not exhaustively evaluated.
- A preregistration natural-language hypothesis phrasing differed from the registered formal estimand; results are reported and interpreted with respect to the archived formal estimand.
- The preregistered benign-control (M = 50) and false-positive-blocking secondary estimand were not executed; therefore no empirical false-positive-blocking claims are made.

VII. CONCLUSION

A preregistered, artifact-backed paired experiment on a synthetic adversarial intent bank found that template-based input sanitization combined with deterministic verifier-gating and at-most-one automated repair produced a small but

statistically detectable absolute increase in deterministic-validator-accepted LLM-generated configurations under the archived formal estimand (paired difference ≈ 0.04255 ; $p = 0.03125$; 95% CI [0.01418, 0.07801]). Diagnostics show the six discordant pairs were associated with permitted repair calls that enabled acceptance. Importantly, this formal-estimand improvement does not equate to measured reduction in attacker success under attacker-centric definitions. The run is limited by synthetic scope, a single model/validator, and a protocol deviation (unset runtime temperature). Future work should execute the preregistered benign-control, measure attacker-centric estimands with an independent semantic-safety oracle, diversify models/validators, and set explicit sampling parameters for reproducible deterministic runs. All principal numbers and reconciliation claims above are supported by archived execution and analysis artifacts referenced in Methods and Results (e.g., execution/task_manifest.jsonl; execution/responses/task-000001-shared-initial.txt; execution/scoring/task-000001-baseline.json; execution/scoring/task-000001-guarded.json; analysis/paired_contingency_table.csv; analysis/condition_summary.csv).

DISCLOSURE STATEMENT

This study was executed autonomously after an autonomy lock under the archived master prompt (provenance/master_prompt.txt, SHA-256: 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582bb39b15ba).

Human role and autonomy boundary: a human Research Director selected the topic family (Generative AI / LLMs for NetOps) and constrained the initial master prompt prior to the autonomy lock; all literature review, preregistration, study selection, code generation, experiment execution, analysis, peer-review cycles, manuscript drafting, and revision reported here were performed autonomously after the lock. Declared models/tools and orchestration: primary generator = gpt-5-mini (execution/model_configuration.json), deterministic validator = deterministic_netops_validator_v5, task generator = deterministic_netops_task_generator_v5, orchestration adapter family = hosted_netops_gvr_v1. Preregistration: study cand-2026-03 was preregistered and archived prior to observing outcomes (preregistration_sha256: ba6ed25e84c7240fbe94f2d50b3e778fbfffd810ff2db632df8946e85f0f1373).

Protocol deviation: the archived run recorded an unset runtime sampling temperature (temperature = null in execution/model_configuration.json); this is disclosed as a deviation because provider-side defaults may affect sampling determinism. To preserve paired comparability the pipeline used shared_initial_candidate semantics (single shared generation per pair) and provider-call accounting documented in analysis/deterministic_reconciliation.json confirms the actual call pattern (shared_initial_generation = 150; repair = 6). Key archived artifact references (examples): execution/task_manifest.jsonl; execution/responses/task-000001-shared-initial.txt; execution/scoring/task-000001-baseline.json; execution/scoring/task-000001-

guarded.json; analysis/deterministic_reconciliation.json;
analysis/paired_contingency_table.csv; analy-
sis/condition_summary.csv; analysis/analysis_log.jsonl;
execution/model_configuration.json; execu-
tion/execution_manifest.json. No human manually edited
or corrected the manuscript technical content after the
autonomy lock.

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